

Rajat Savdekar

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EDUCATION

University of Southern California, Los Angeles Master of Science, Computer Science Coursework: Algorithms, Database Systems, Machine Learning, Web Technology, VR / XR Development	Aug 2024 – May 2026 GPA: 3.63
University of Mumbai, Mumbai Bachelor of Engineering, Computer Engineering Coursework: Operating Systems, Computer Networks, Distributed Systems, Natural Language Processing	Aug 2019 – May 2023 GPA: 3.7

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C++ , HTML/CSS
Backend & APIs: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, Server-Sent Events, Microservices
Frontend: React, Next.js, Tailwind CSS
AI & ML: LLMs, RAG, AI Agents, LLM Evaluation, NLP, Model Quantization, PyTorch, OpenAI API, Whisper
Data & Storage: PostgreSQL, MySQL, MongoDB, Redis, Prisma, LevelDB, Yjs (CRDTs), Object Storage
Cloud & Infrastructure: AWS, Docker, Terraform, GitHub Actions, CI/CD, Linux, Git

PROFESSIONAL EXPERIENCE

Software Developer Easley-Dunn Productions, Inc. - Reduced AI narration inference cost by 60% and replay turnaround from 22 to 4 minutes by implementing a warm 4-bit quantized 7B worker, batched speech synthesis, and content-hash caching. - Built a human-in-the-loop grounding gate for AI surgical narration, eliminating ungrounded operative notes by validating each generated sentence against the source event log. - Raised replay upload success from 82% to 99.6% across 150+ sessions by replacing 100 MB in-memory SQLite blobs with streamed object storage and Postgres metadata.	Aug 2026 – Present Torrance, CA
Software Engineer Jalgaon Fruit Sales Cooperative Ltd – Agritech Supply Chain Platform - Decomposed a legacy PHP monolith into Python Flask microservices backed by PostgreSQL with cron-based state sync, cutting query latency by 40% across 1,000+ daily truck transactions. - Doubled peak harvest write throughput by architecting an isolated write path that eliminated 10–15 minute MyISAM lock freezes, sustaining a season where platform volume grew from \$100K to \$200K. - Delivered a web procurement platform with REST APIs and live analytics dashboards, migrating 150+ daily clerks across 5 warehouses off manual logs.	May 2023 – May 2024 Jalgaon, India

SELECTED PROJECTS

MockPad – Real-Time Collaborative Code Editor - Architected a real-time collaborative editor on Yjs CRDTs over custom WebSockets, sustaining sub-100ms sync latency for 30 real active users on a single \$6/month DigitalOcean droplet. - Cut server peak memory consumption by 80% with zero session data loss by implementing an LRU disk-eviction strategy to LevelDB for inactive collaborative rooms.	[Live] [GitHub]
Caliberate – LLM Code-Eval Harness - Built a sandboxed eval harness executing LLM-generated patches across parallel network-isolated Docker workers, holding zero task loss through a PostgreSQL SKIP LOCKED queue with lease-based crash recovery. - Calibrated an LLM-as-judge to 0.81 agreement against 150 hand-labeled patches, then gated CI so a 5-point pass-rate regression on the golden set blocks merge.	[GitHub]
CloudBridge – AI Cloud Architecture Advisor & Topology Engine - Engineered a FastAPI synthesis engine turning plain-language requirements into validated AWS topologies, blocking designs that violate Well-Architected pillars or IAM least-privilege before Terraform export. - Built a React Flow topology engine recalculating multi-service dependencies and cost projections against a pinned 12-service AWS price catalog on every parameter change.	[GitHub]

PUBLICATIONS & HACKATHONS

Research Publications. <i>Texture Feature Analysis Using GLCM Matrix (10 citations).</i>	[GitHub]
Stitch – Autonomous CI Repair Agent. Aug 2026. OpenAI Build Week, Developer Tools track.	[GitHub]
Aura – visionOS Accessibility Assistant. Aug 2025. 2nd place of 24-hour hackathon.a	[Demo]